

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element D Initial Template

Element D: Design Concept Generation, Analysis, and Selection

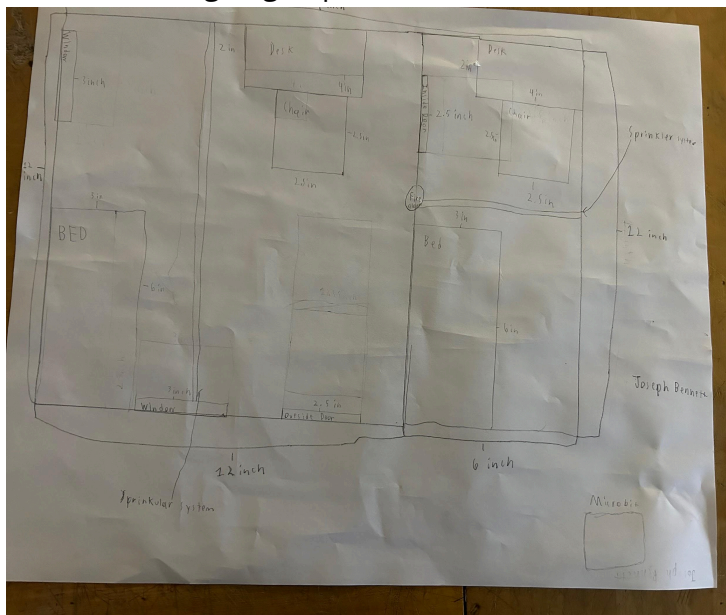
Design Idea Generation (D1)

Design Idea #1

Date: 3/11/25

Description: a layout of what the floor would look like. Also where the suppression system would hang/lay. This included the rooms and major pieces of furniture. It also didn't have that many details down other than the measurements that were 18x12in (With two rooms, one being 12in and the other being 6in)

Since it was only a concept at first, there are no materials labeled, this is because our instructor was going to provide the selected materials that would be used,



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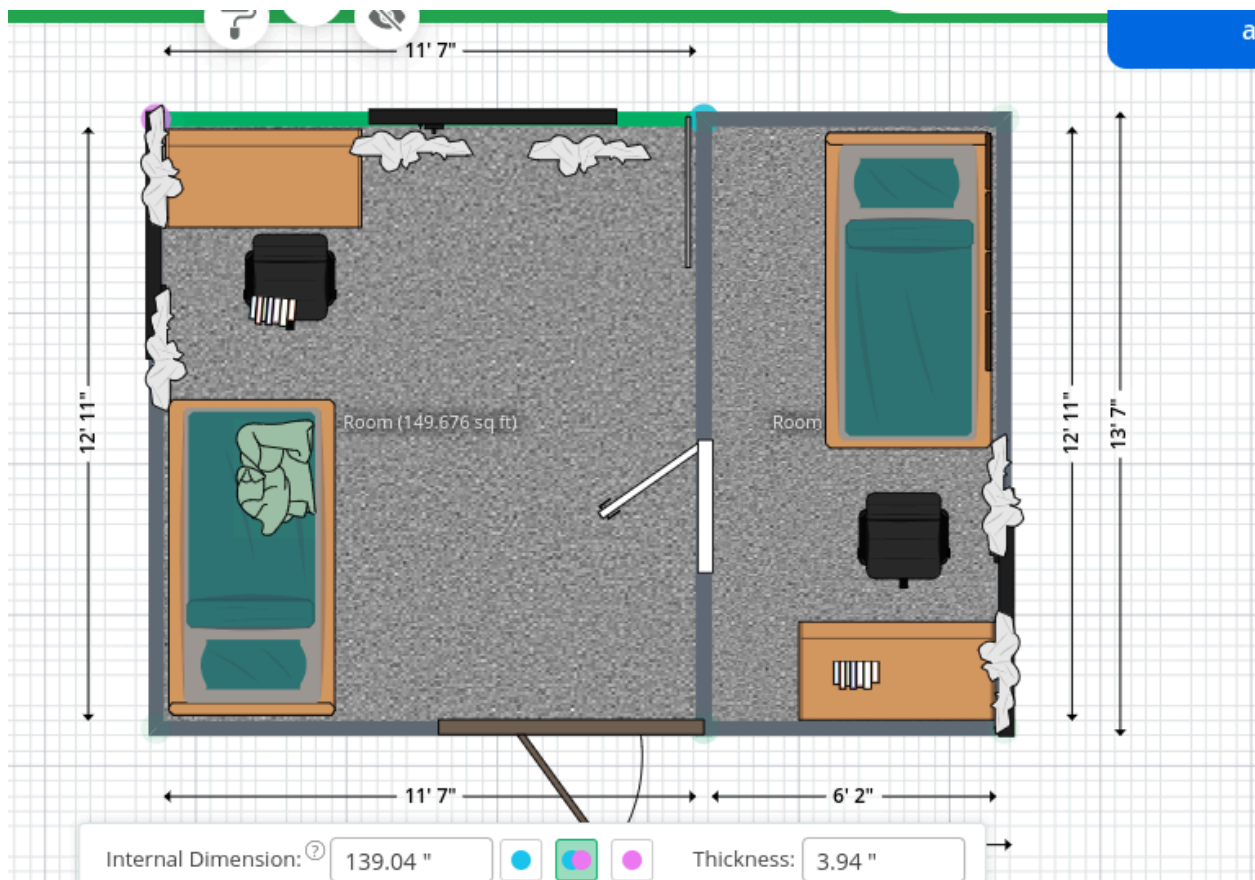
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Design Idea

Date: 3/13/25

Description: This model was made digitally, using the same exact measurements as all of them, to see if it would look good as a real room. This model was made in *Planner 5D* and does not include the suppression system because we were only planning out the room. This model included a little more detail of what material would look like.



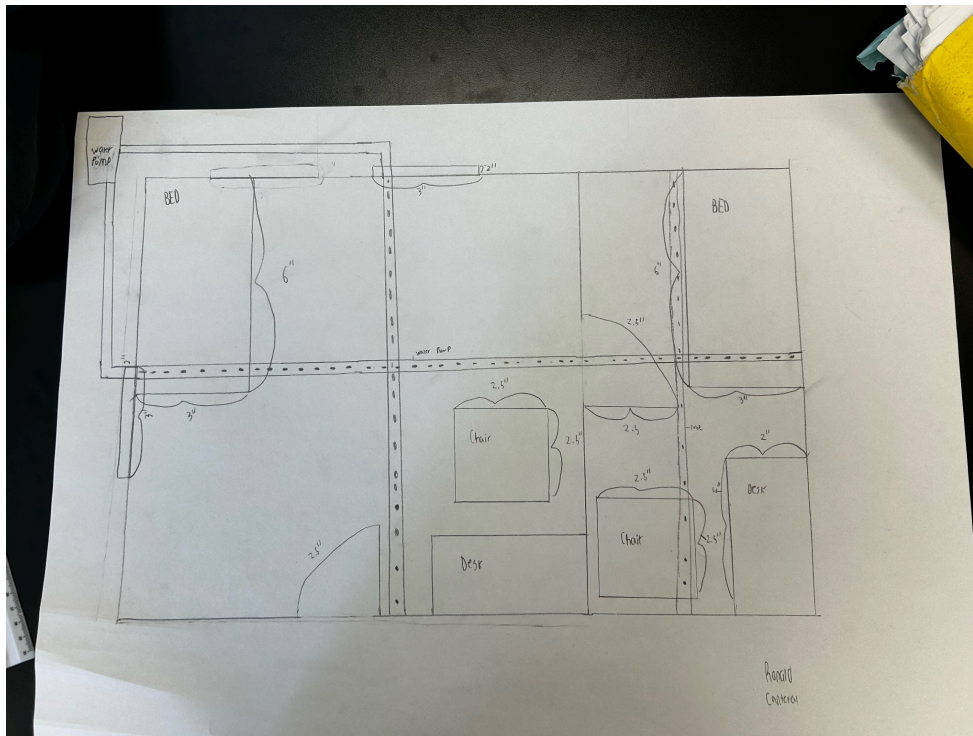
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Date: 3/18/25

Description: We realized that the others weren't the best ideas. And instead we went with another one that our teammate came up with. This one has a better layout for the suppression system that is shaped like a "H" in order to sprinkle through both of the rooms. This design has a better overall suppression system (Drawn, and the idea in its entirety.) Again, there were no materials listed due to our materials being limited in choice and we were unsure in advance of what would be supplied by our instructor.. 879



Design Idea Comparison and Selection (D1, D2)

	weight	Option 1	Option 2	Option 3
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Cost	1-5	3	3	3
Physical durability	1-5	2	1	5
Effectivity	1-5	2	1	3
Design	1-5	1	3	3
Time (+least, -most)	1-5	3	5	4
score		11	13	18

Our team chooses design option 3 to solve the community problem.

Design Blueprint (D4)

Design Blueprint #1

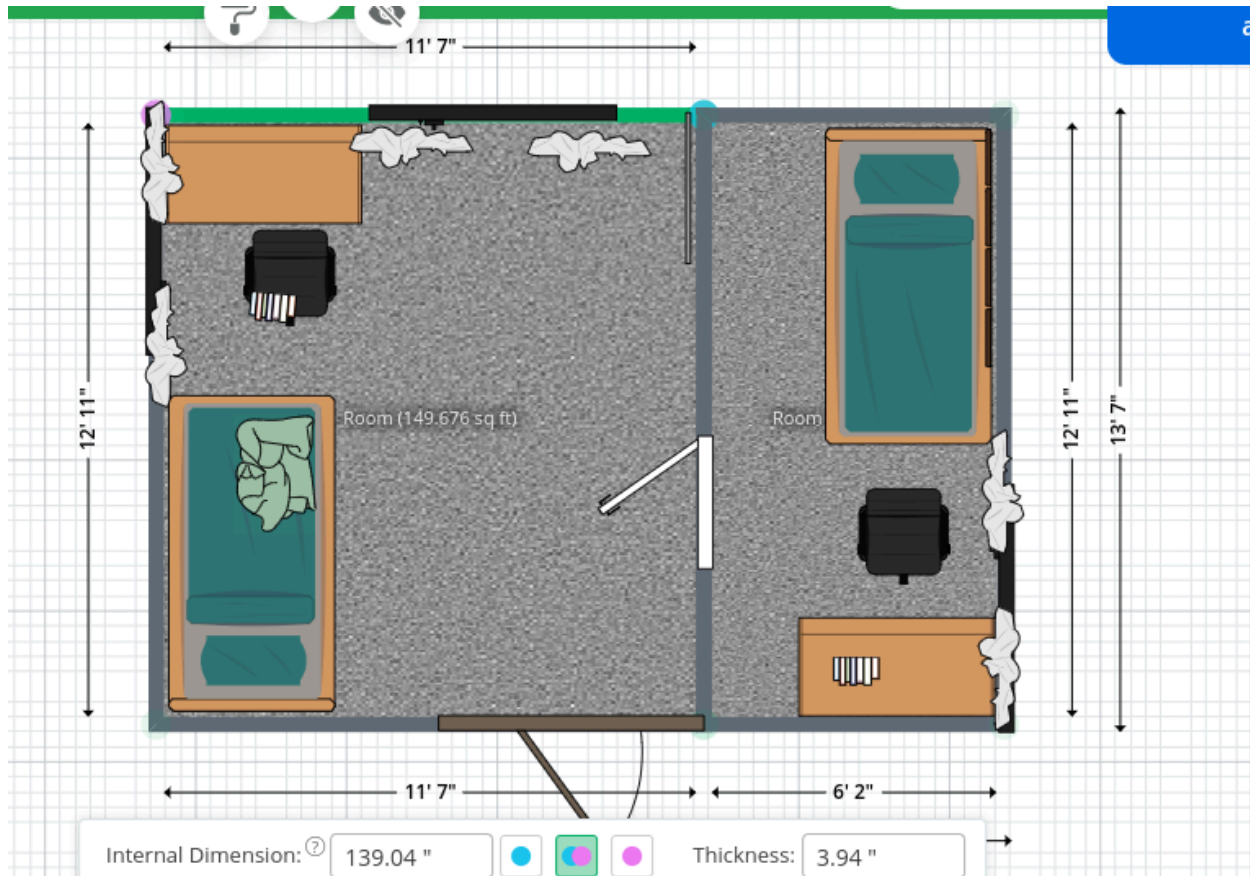
Date: 2/21/25

Sketch:

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Manufacturing: the floor and walls/ overall structure will be made from cardboard. This is just a design, and testing before making the real thing so everything will be held together with glue. So will the furniture.

Step 1: Cutting out all pieces of cardboard that will be needed. The measurements of the walls that would be cut were 18 x 12 inches. And the holes for the door were also cut 6 x 2.5 Inches. The ceiling and floor had the same measurements. Then a window would be cut out on two of the wall panels of cardboard (2 x 3 inches.) Then it was time to cut out the pieces of cardboard for the furniture. (Note that all of the furniture had individually cut panels for height and width/length to make cubes or rectangular prisms to mimic how much space the furniture would take up.) The beds measured 6 x 3 inches. The tables were 2 x 4 inches. And the chairs were 2.5 x 2.5 inches.

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Step 2: Glueing everything together. The walls would be glued to the floor to make a rectangular prism missing one side; the ceiling. The ceiling had to be removable to be able to see the inside and place all the furniture in there once everything was complete. It was held together with hot glue. After it was stuck together, we then planned out where the furniture would go.

Step 3: Once we had everything in place, we made the curtains that would go on the window cut outs. We cut half a popsicle stick (Cut horizontally.) Then took some fabric and cut holes along the top to stick the popsicle stick into. And that was then glued above the window with hot glue. Even though we should have, we didn't measure the exact measurements of fabric used, we cut the length according to where the fabric would touch the floor, and started just above the windows.

Supply List and Potential Costs:

Item	Cost each	# needed	Total
Cardboard	1.65\$	6	9.90\$
Hot glue stick	0.09\$	5	0.45\$
Scissors	18\$	2	36\$
Total Cost	19.74\$	13	46.35\$

Plan of Action (D3)

First we brainstormed ideas. Then each member of our group made a different sketch and we picked the best sketch. We then improved on that sketch to make it better.

Iterations for the Solution: Rearranged the house

Design Blueprint #2

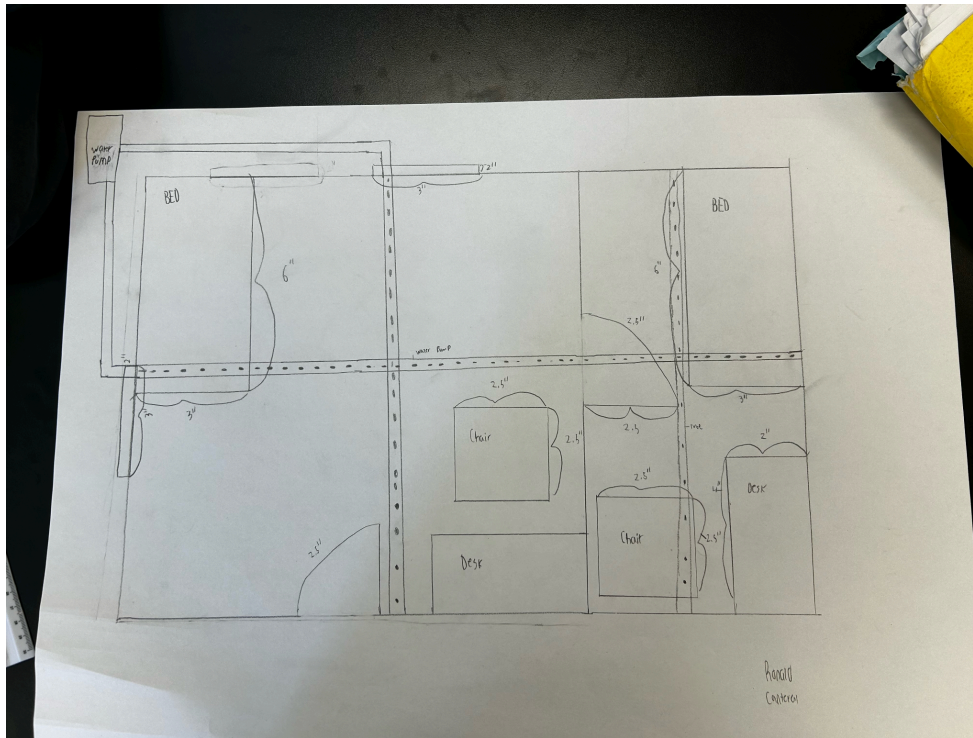
Date:

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Sketch:



Manufacturing Methods:

Step 1: First we collected the 18 inch 10-series 80/20 aluminum bars with tapped ends (4), 10 inch 10-series 80/20 aluminum bars (8), 12 x 18 tempered glass pane (1), 12 x 18 x 0.134 hardboard roof and back wall panels (2), 12 x 12 x 0.134 hardboard end wall panels (2), 10.5 x 16.5 x 0.134 hardboard floor panel (1), 11.5 x 12 x 0.134 hardboard interior wall panel (1), 80/20 part 33440 inside-inside hidden corner connectors (16), 80/20 part 4108 2-hole lite inside corner bracket (4), 80/20 part 3390 1/4-20 x 0.375 button head bolt (4), 80/20 part 3382 1/4-20 slide in economy t-nut (4), 1/4-20 button head bolt and nut (12), 1/4-20 button head nut (4), 1x1 hook and loop adhesive fastener (4).

Step 2: We then measured the wood and cut the wood out for the doors and windows of the house using a 16-inch variable-speed scroll saw and a ruler and drilled holes into the hardboard using a drill with a 5/16 inch drill bit. We then built our aluminum bars 18 inch 10-series 80/20 aluminum bars with tapped ends (4) and 10 inch 10-series 80/20 aluminum bars (8)) using the 80/20 part 33440 inside-inside hidden corner connectors (16) connect each beam to each other as the base of our apartment. We then created the furniture for the apartment and created the detection system for the apartment.

Step 3: Finally we installed the walls of the house for the apartment by using the 1/4-20

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button head bolt and nut (12) and 1/4-20 button head nut (4) using a 5/32 inch hex key to screw in the nuts and bolts to keep the hardboard of the walls in place.

Supply List and Potential Costs:

Item	Cost each	# needed	Total
Hardboard walls	13.59\$	6	81.54
Aluminum bars	10.81\$	12	129.72
Corner connectors	6.83	16	109.28
Total Cost	31.23	34	320.54

Rubric

Element D: Design Concept Generation, Analysis and Selection						
	5	4	3	2	1	0
1. Process for Generating & Comparing Possible Solutions. The Process for Generating Possible Solutions _____.	was <i>comprehensive, iterative, and consistently</i> defensible (viable design highly likely)	was <i>thorough, iterative, and</i> generally defensible (viable design likely)	was <i>adequate, generally</i> iterative and defensible (<i>viable design is possible</i>)	was <i>partial or overly general and only somewhat</i> iterative and/or defensible (raising issues with the design viability)	was <i>incomplete or minimally iterative and/or defensible</i> (raising issues with the design viability)	was not evident that there was an attempt



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2.Design Solution Choice. The Design Solution chosen _____	<i>was well-justified and demonstrated attention to all design requirements</i>	<i>was justified and demonstrated attention to most if not all design requirements</i>	<i>was explained with reference to at least some design requirements</i>	<i>was not sufficiently explained with reference to design requirements</i>	<i>lacked support related to design requirements</i>	<i>was not attempted or evidence not provided</i>
3. Plan of Action. The plan of action _____	<i>has considerable merit and would easily support repetition and testing for effectiveness by others</i>	<i>would support repetition and testing for effectiveness by others</i>	<i>might not clearly or fully support repetition and testing for effectiveness by others</i>	<i>has insufficient detail to allow for testing for replication of results</i>	<i>is lacking to the point that the design solution cannot be tested</i>	<i>cannot be executed or does not exist</i>

